System Design

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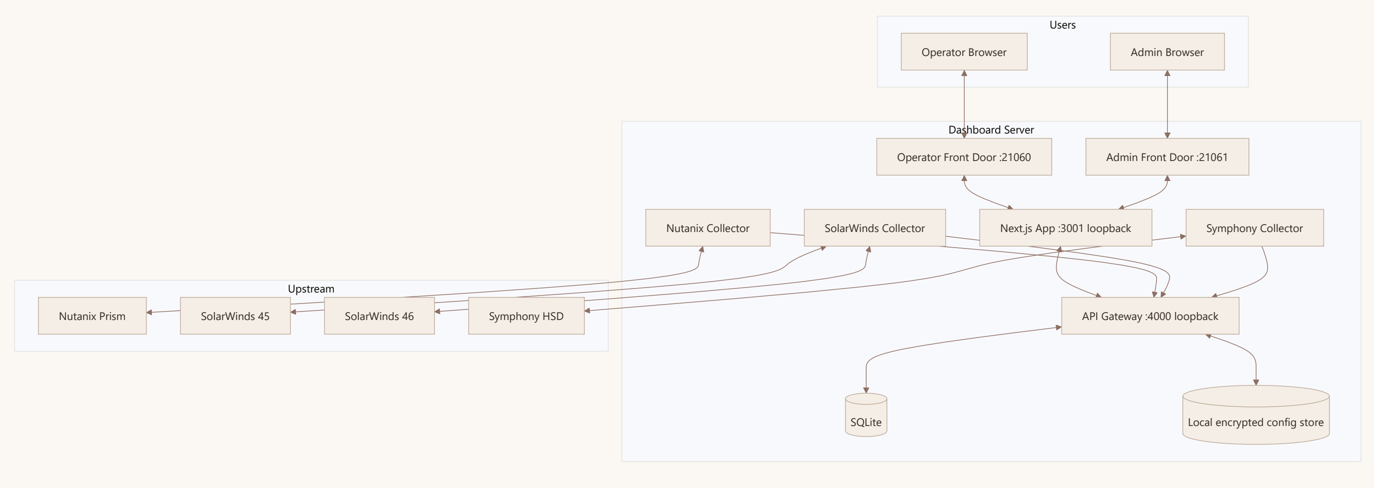
1. Purpose

Describe the implemented runtime architecture, trust boundaries, data flow, deployment topology, and recovery model for the UAIL IT Dashboard.

2. System Context

The system is a multi-process Node.js application deployed on a Windows host. It presents separate operator and admin surfaces while sharing the same internal application stack.

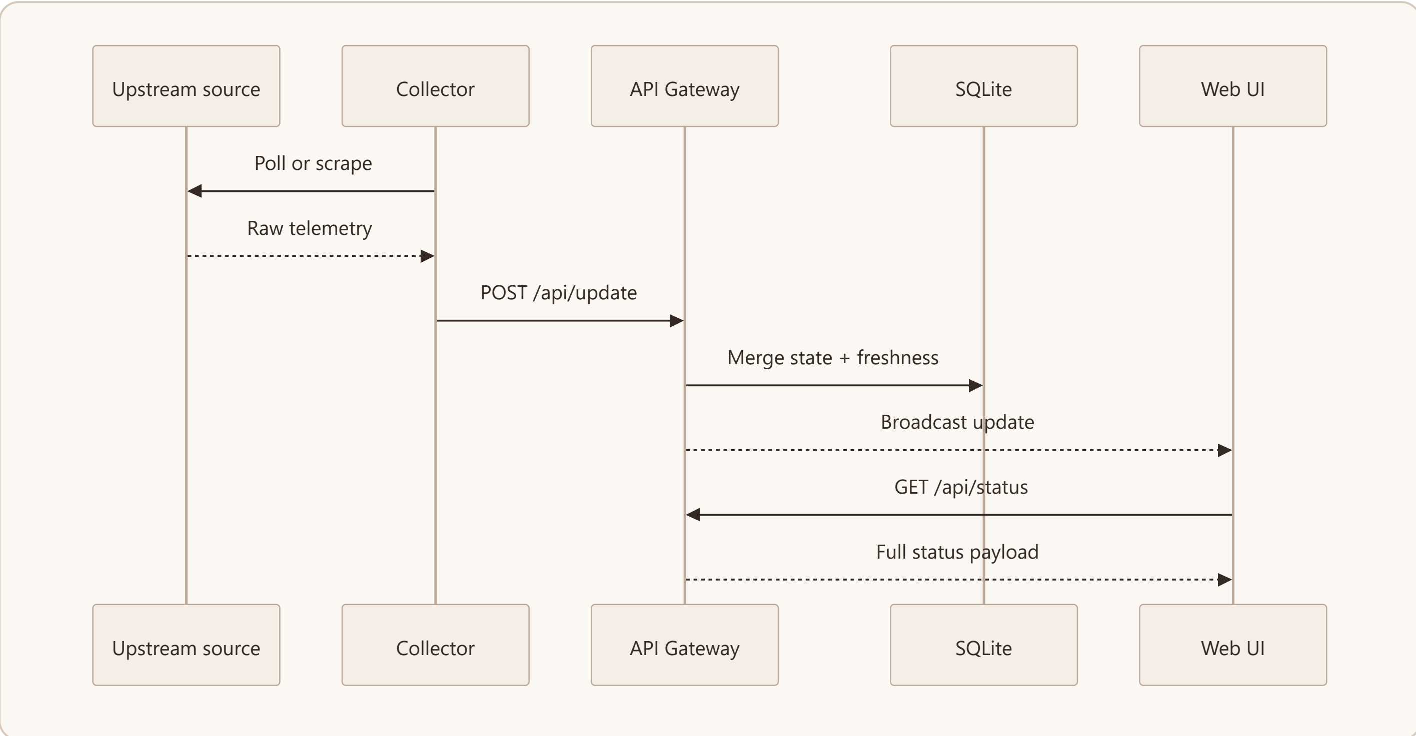


3. Runtime Components

Component	Purpose	Exposure
`dashboard-frontdoor-operator`	Reverse proxy and operator surface identity	LAN
`dashboard-frontdoor-admin`	Reverse proxy and admin surface identity	LAN

`dashboard-ui`	Shared Next.js web app	Loopback
`api-gateway`	REST API, WebSocket broadcast, admin endpoints	Loopback
`nutanix-collector`	Direct Nutanix API polling	Host-local
`solarwinds-collector`	SolarWinds portal scraping for servers and networks with independently configured portal credentials	Host-local
`symphony-collector`	Symphony HSD portal scraping	Host-local

4. Data Flow



5. Dashboard Domains

5.1 HCI Domain

- Source: Nutanix
- Primary content: cluster CPU, memory, storage, node state
- Trust rule: Nutanix-only

5.2 HSD Domain

- Source: Symphony HSD
- Primary content: open work items, SLA posture, special queues
- Trust rule: last-success state remains visible if collection fails

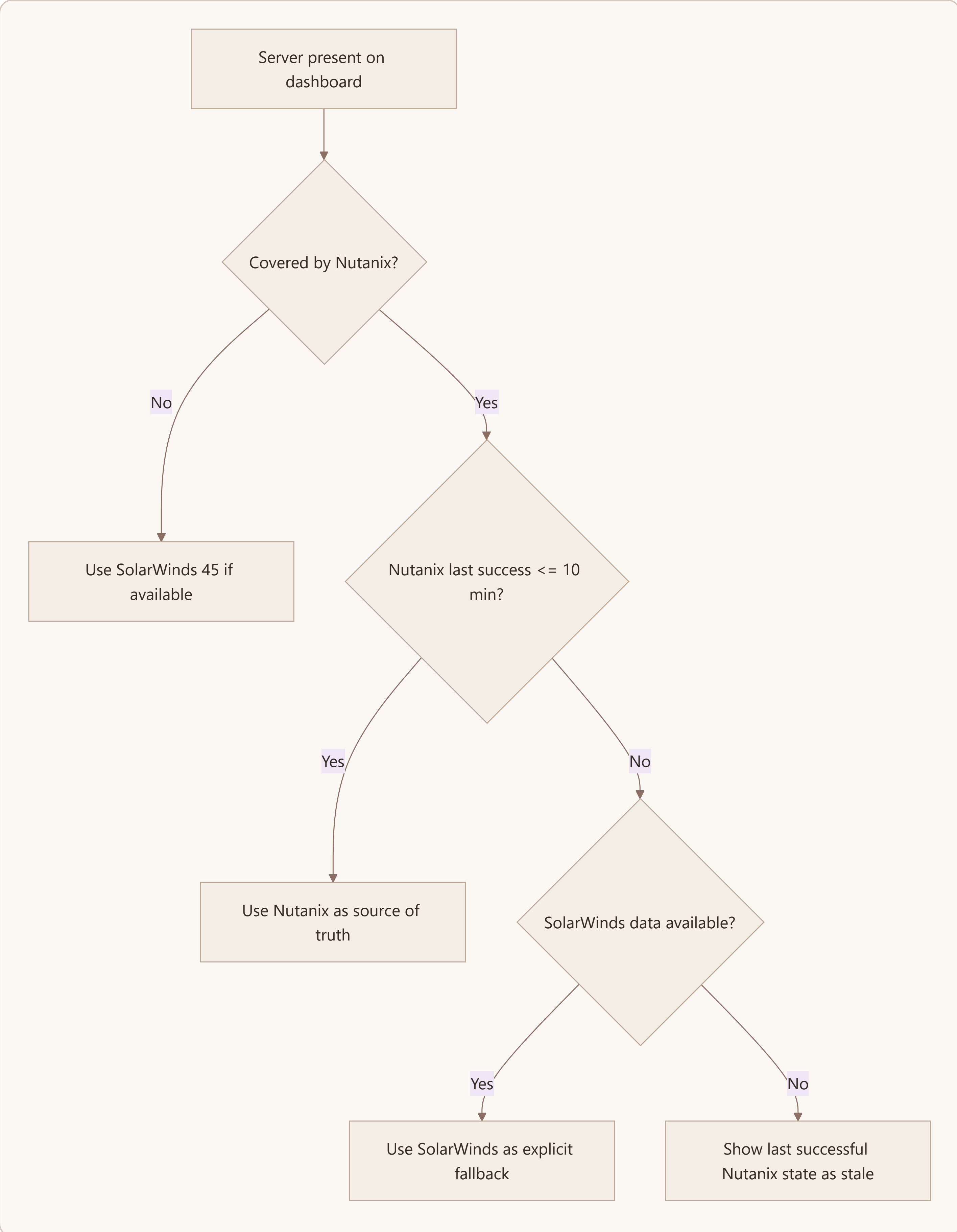
5.3 Network Domain

- Source: SolarWinds 46
- Primary content: carrier and SDWAN state, Tx/Rx, sparklines
- Trust rule: SolarWinds 46 is authoritative

5.4 Server Domain

- Primary sources:
  - Nutanix for HCI-backed servers
  - SolarWinds 45 for on-prem servers
- Fallback rule:
  - for overlapping coverage, Nutanix remains primary
  - SolarWinds is used only after Nutanix is stale for more than 10 minutes and SolarWinds data exists

6. Source Of Truth Decision Flow



7. Freshness And Failure Model

Each section is represented with operational freshness metadata:

- poll interval
- last attempt time
- last success time
- last error

The UI uses this data to derive link state and staleness messaging. The dashboard does not fabricate values; it reuses the last confirmed values and makes the freshness visible.

8. Storage Model

8.1 Primary Runtime Store

- SQLite
- current merged dashboard state
- freshness and error state

8.2 Local Config And Secrets

- runtime config and secrets stored under the shared runtime root
- encrypted local secret handling supported through the configured secret-store passphrase
- installer and bootstrap paths maintain separate credentials for Nutanix, SolarWinds 45, SolarWinds 46, and HSD

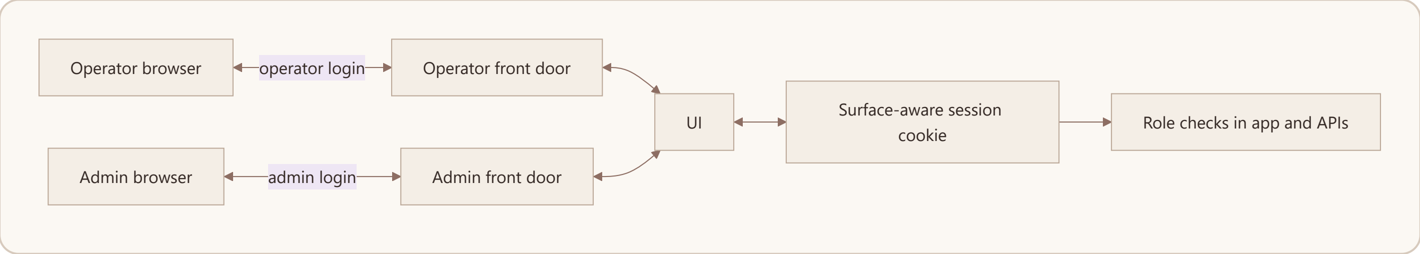
8.3 Local Audit Trail

- admin and authentication audit records are stored as append-only JSONL files under `C:\ProgramData\UAIL\ITDashboard\audit`
- the admin Audit lane and export actions read from this local store
- this audit path is independent of PostgreSQL and is part of the supported baseline

8.4 Optional Extensions

- PostgreSQL-related source code remains in the repository for future maturity work
- PostgreSQL is not part of the supported installer or deployment baseline for the current release

9. Authentication And Surface Separation



- operator and admin surfaces have different login prompts
- both surfaces share the same Next.js application
- the surface determines the expected role and cookie namespace
- admin APIs are restricted to admin sessions

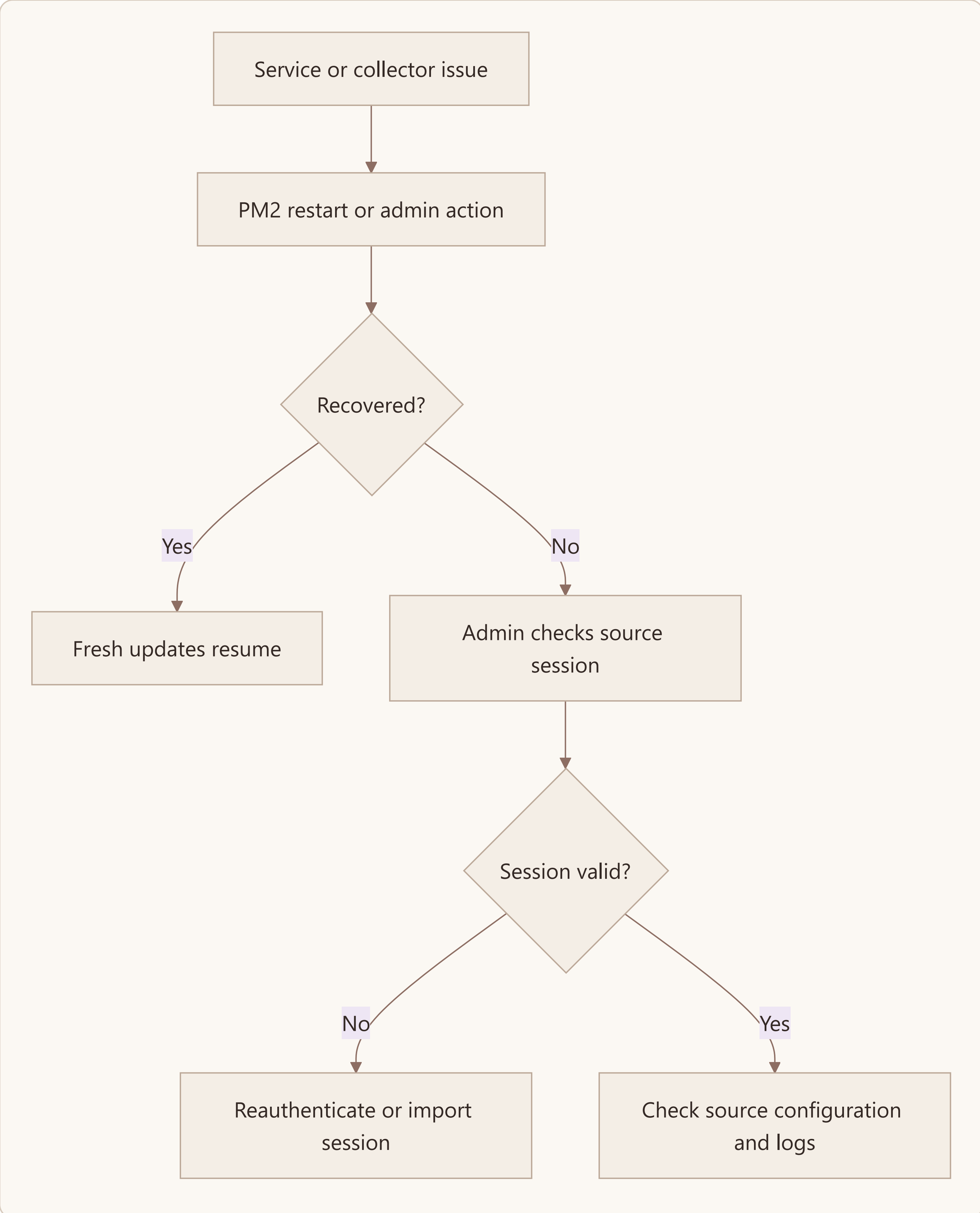
10. Deployment Design

Item	Value
Primary host model	Dedicated Windows server or VM
Operator port	`21060`
Admin port	`21061`
Internal UI port	`3001` loopback
Internal gateway port	`4000` loopback
Process manager	PM2
Runtime store	SQLite
Audit store	Local append-only JSONL files under runtime root

Bootstrap expectations:

- installer and staged deployment prompt separately for SolarWinds 45 credentials, SolarWinds 46 credentials, and HSD credentials
- generated runtime configuration uses `SW_SERVERS_*`, `SW_NETWORKS_*`, and `SYM_*` variables rather than assuming a shared portal identity

11. Auto-Heal And Recovery



Auto-heal mechanisms:

- PM2 process supervision
- saved process list via `pm2 save`
- Windows startup recovery via a SYSTEM scheduled task that reruns the bundled PM2 bootstrap
- runtime permission repair for PM2 state, session files, logs, config, and app data before bootstrap
- admin-triggered restart actions
- browser wake-lock request on the operator surface when the browser supports secure-context screen wake lock

Screen-display note:

- the operator UI attempts to keep the screen awake through the browser when supported
- Windows lock-screen, screensaver, or group-policy enforcement can still override browser behavior, so dedicated wallboard machines should still be configured appropriately at the OS level

12. Security Boundaries

- only the two front-door ports should be LAN-exposed
- UI and gateway remain loopback-only
- source credentials and session state are sensitive operational assets
- the dashboard is intended for internal network deployment only

13. Key Constraints

- Nutanix TLS verification is relaxed because of upstream certificate conditions
- HSD and SolarWinds depend on browser-authenticated sessions
- current authentication is practical and local, not enterprise SSO

14. Design Principles

- show only real data from a trusted source
- make freshness explicit
- prefer operational simplicity over avoidable platform complexity
- keep admin operations available through the web surface rather than requiring shell access for normal recovery